

Adding time-constant component hazard ratios mispredicted a combination's RMST when effects were delayed or follow-up differed

A simulation study answering catalog problem CMP-18

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2026-09-23

Abstract

Background. Component survival models add time-constant log hazard ratios. A time-constant fit to a time-varying effect is a least-false average weighted by each trial's own event times, so two components' averages from different trials need not add up to the combination's. Catalog problem CMP-18 asks whether this matters for the estimands decisions use.

Methods. Trials of each of two components against control, 300 per arm, with constant, jointly delayed, or delayed-versus-waning effects, and follow-up of 3 years in both or 1.5 and 3 years; stratified Cox models with time-constant or piecewise (before and after one year) component effects predicting the combination's 3-year RMST difference; bootstrap SEs; 500 replicates per cell.

Results. With both effects delayed, the time-constant prediction was biased by 0.055 years (truth 0.405), coverage 0.884, and the piecewise model reduced the bias to 0.017 (coverage 0.942). With delayed and waning effects and unequal follow-up, the time-constant prediction was biased by -0.102 (coverage 0.820) and the piecewise model by -0.072 (coverage 0.906), because neither can learn a late effect the shorter trial never observed.

Conclusion. A time-constant component effect is adequate only when the combination's effect is itself roughly constant and the trials cover the same follow-up. A time-varying extension fixes the first condition, not the second.

1 The problem

A Cox fit to a non-proportional effect converges to a least-false hazard ratio that depends on the event-time distribution and censoring (1). Component models add such summaries across trials with different follow-up and different effect profiles, and then apply the sum to a combination nobody observed. The prediction is right only if the summaries' weights match and the combined effect is constant in time.

2 Design

Registered protocol: `protocol.md`. Control hazard Weibull (shape 1.2, scale 3 years). Log hazard ratios: constant (-0.4 each), both delayed ($-0.5(1 - e^{-t/0.7})$), or A delayed and B waning ($-0.5e^{-t/0.7}$), whose sum is constant at -0.5 . The combination's log hazard ratio is the sum. Prediction: trial 1's control Nelson-Aalen cumulative hazard times the fitted hazard ratios; SEs from 60 bootstrap resamples within trial arms.

3 Results

Table 1: 3-year RMST difference of the combination (years); 500 replicates per cell, bias MCSE at most 0.005.

component profiles	follow-up A, B (years)	truth	constant: bias (coverage)	piecewise: bias (coverage)
constant	3.0, 3.0	0.481	-0.004 (0.932)	-0.008 (0.940)
same	3.0, 3.0	0.405	0.055 (0.884)	0.017 (0.942)
different	3.0, 3.0	0.327	-0.011 (0.956)	-0.013 (0.952)
different	1.5, 3.0	0.327	-0.102 (0.820)	-0.072 (0.906)

The registered rule returned that neither model was nominal where it mattered (Table 1): at unequal follow-up the piecewise model covered 0.906, below the 0.93 required. Two failures appeared. **Delayed effects:** a constant hazard ratio moves benefit earlier than a delayed one, and RMST credits early benefit, so the time-constant prediction overstated the combination’s gain even with equal follow-up; the piecewise model largely corrected it. **Unequal follow-up:** component A’s effect after 1.5 years was never observed, so both models extrapolated its effect from the early period, where it had not yet appeared. With delayed and waning effects and equal follow-up the time-constant model was accurate, because the two effects summed to a constant and the trials weighted time alike.

4 What this does not answer

Two single-component trials rather than a component network; no disconnection or population adjustment; one piecewise extension; one horizon. Peer review has not been done.

References

1. Christopher A. Struthers, John D. Kalbfleisch. Misspecified proportional hazard models. *Biometrika*. 1986;73(2):363–9. doi:[10.1093/biomet/73.2.363](https://doi.org/10.1093/biomet/73.2.363)